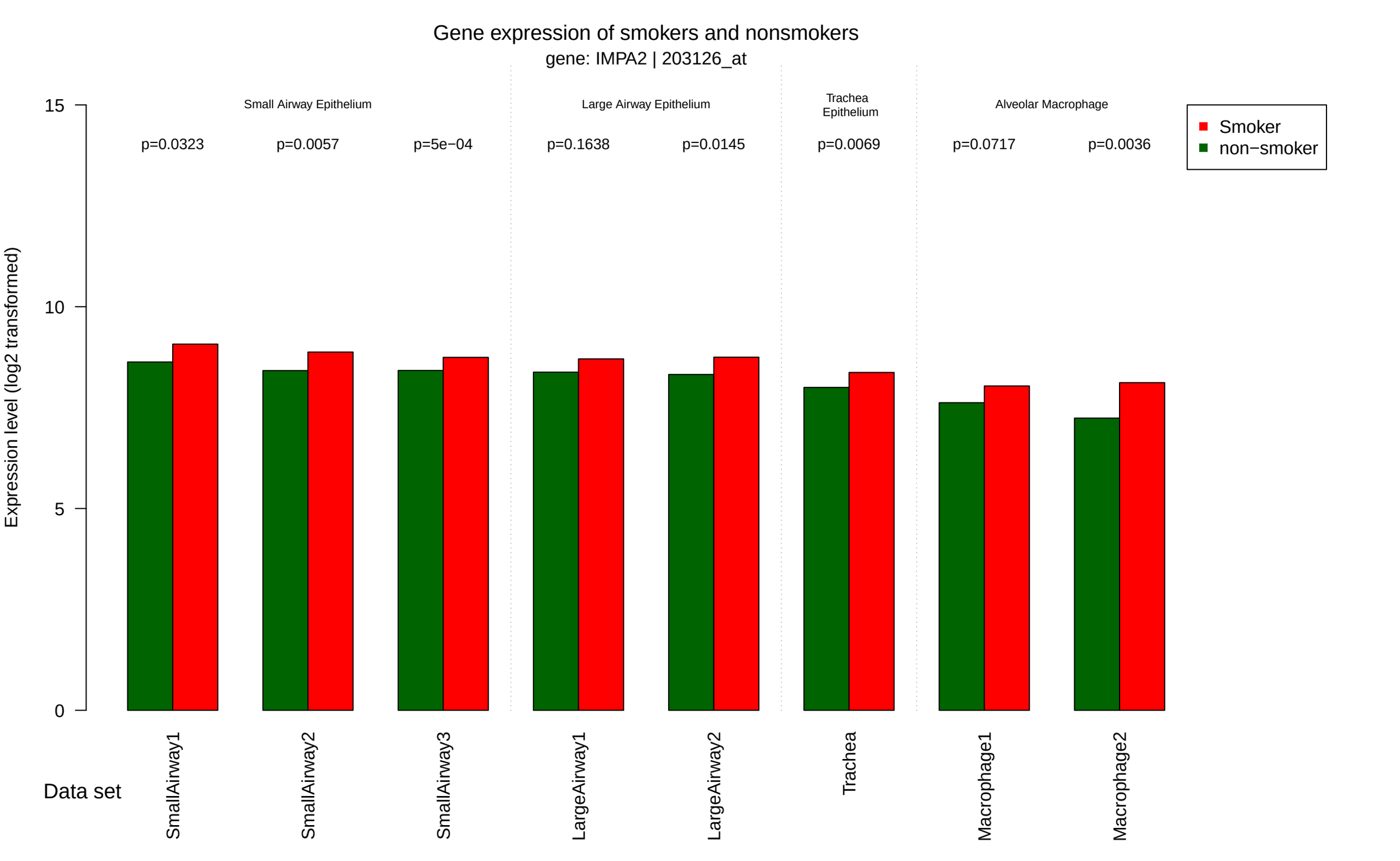
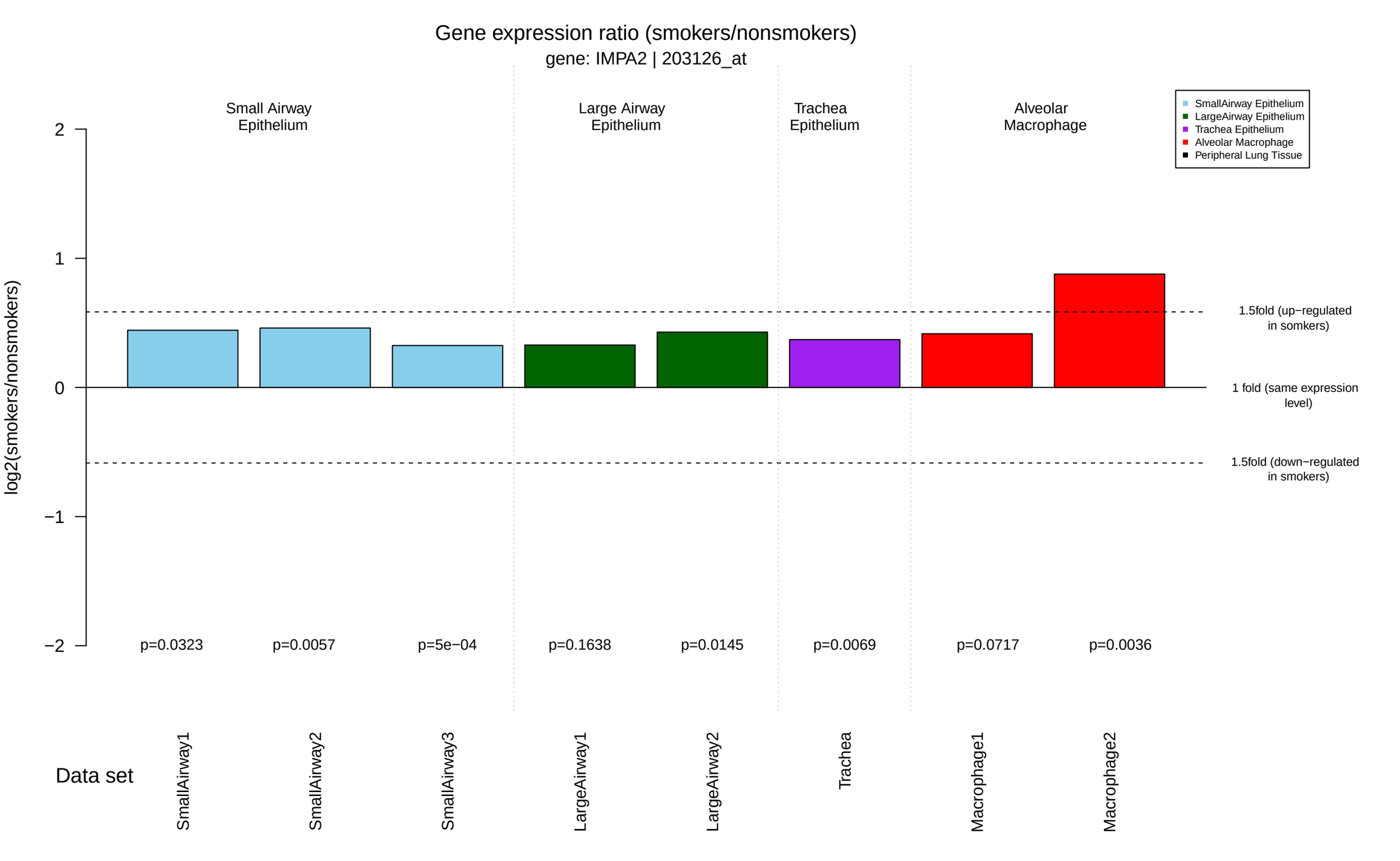
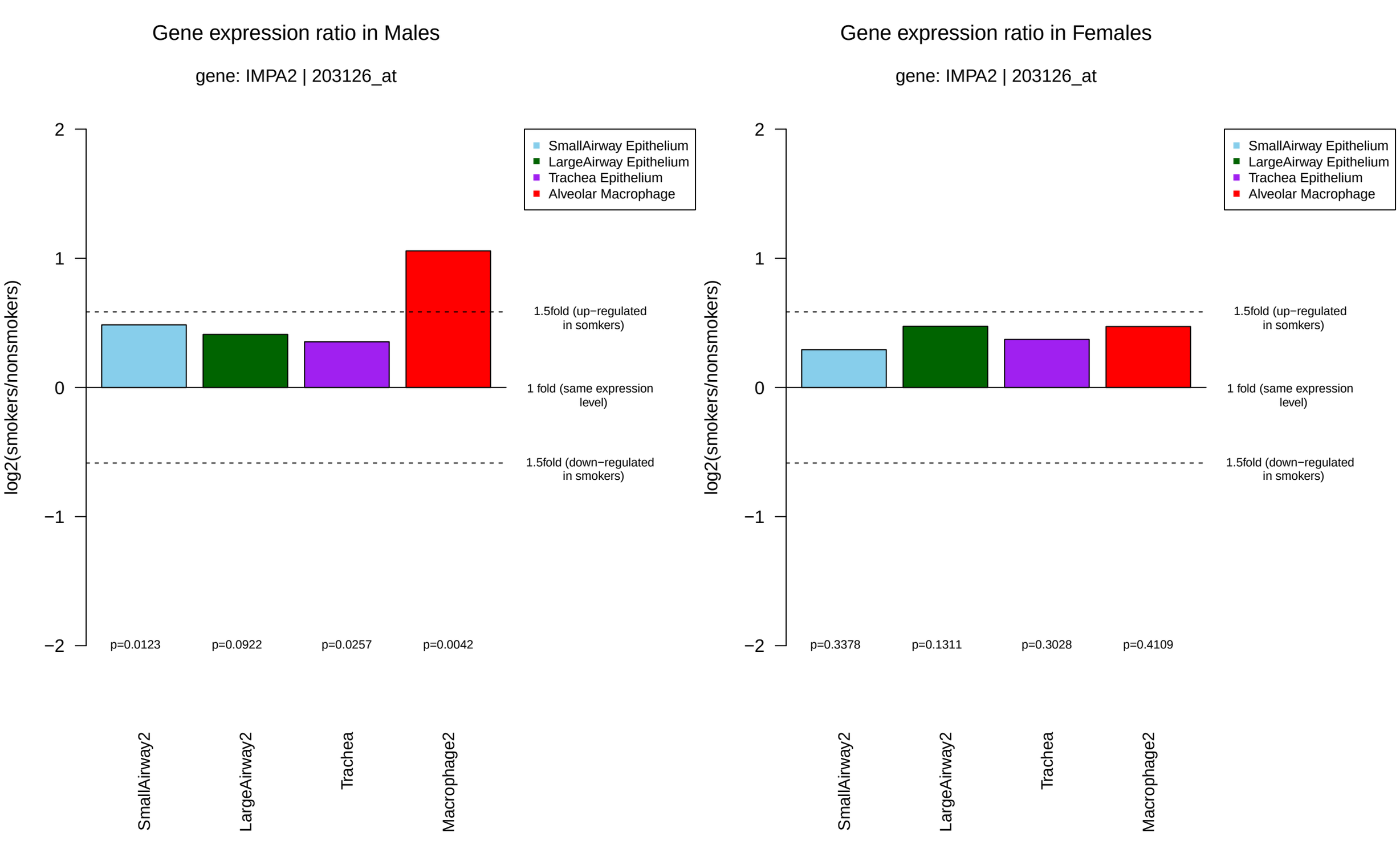
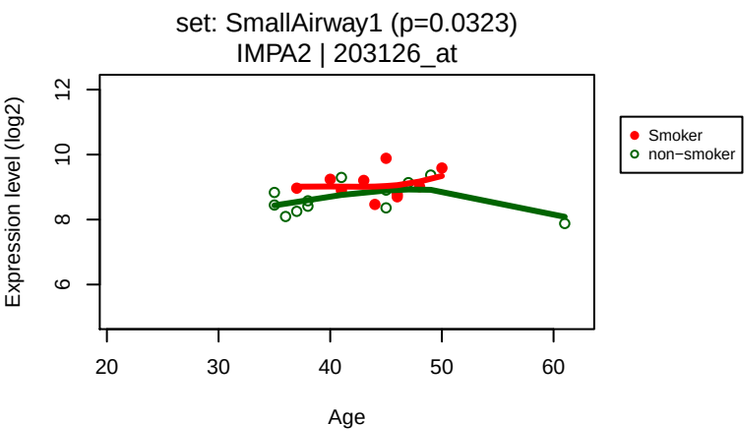
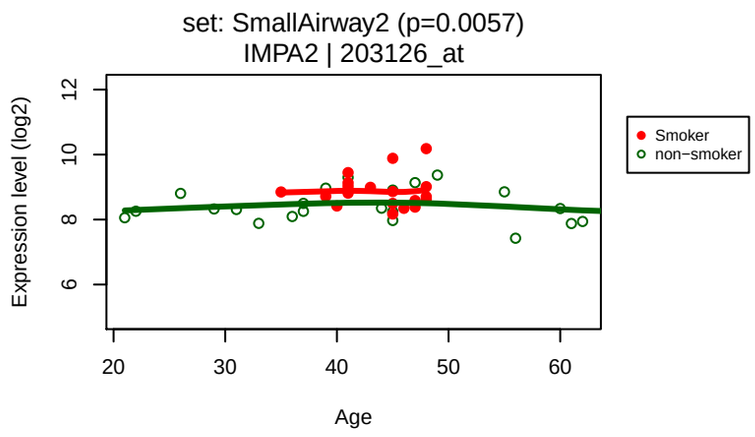
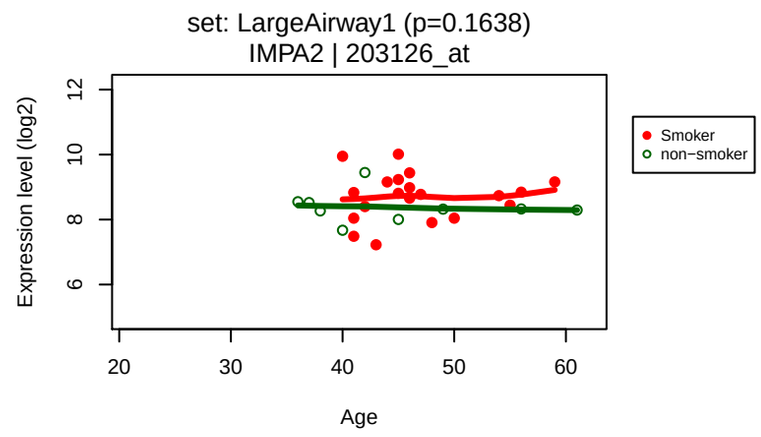

Gene expression of smokers and nonsmokers
gene: IMPA2 | 203126_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0323
p=0.0057
p=5e−04
p=0.1638
p=0.0145
p=0.0069
p=0.0717
p=0.0036
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

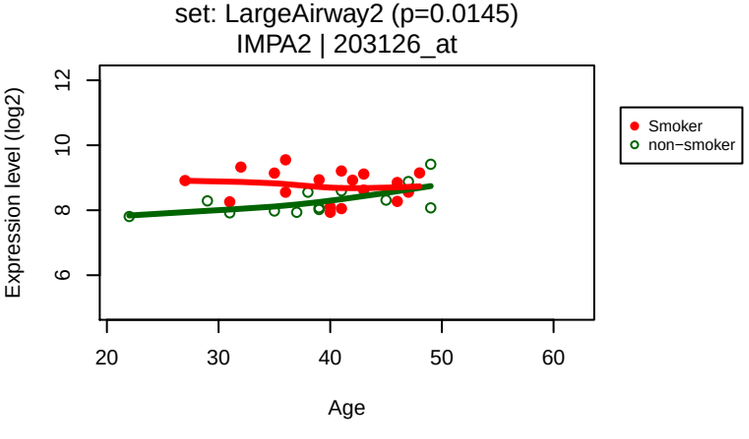

Gene expression ratio (smokers/nonsmokers)
gene: IMPA2 | 203126_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0323
p=0.0057
p=5e−04
p=0.1638
p=0.0145
p=0.0069
p=0.0717
p=0.0036
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

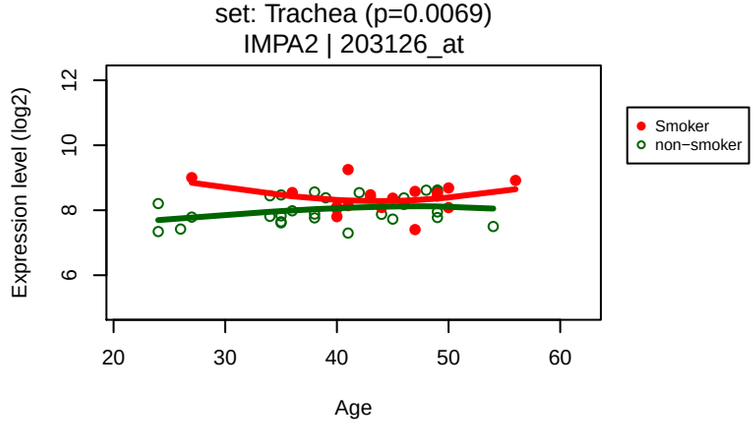

Gene expression ratio in Males
gene: IMPA2 | 203126_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0123
p=0.0922
p=0.0257
p=0.0042
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IMPA2 | 203126_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3378
p=0.1311
p=0.3028
p=0.4109
SmallAirway2
LargeAirway2
Trachea
Macrophage2

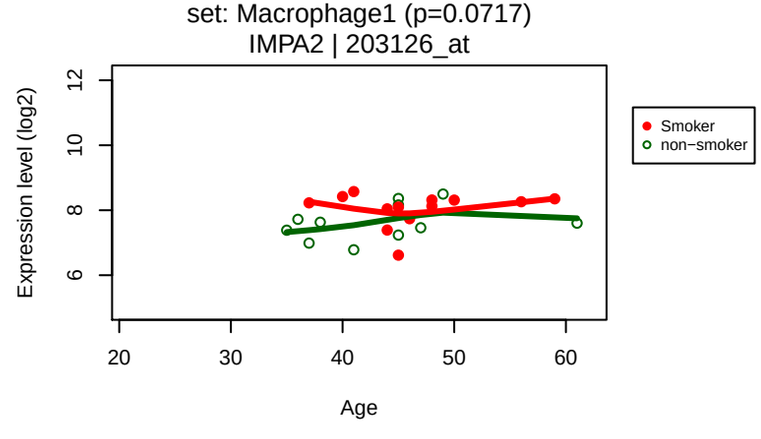

set: SmallAirway1 (p=0.0323)
IMPA2 | 203126_at
Expression level (log2)
Age
Smoker
non−smoker

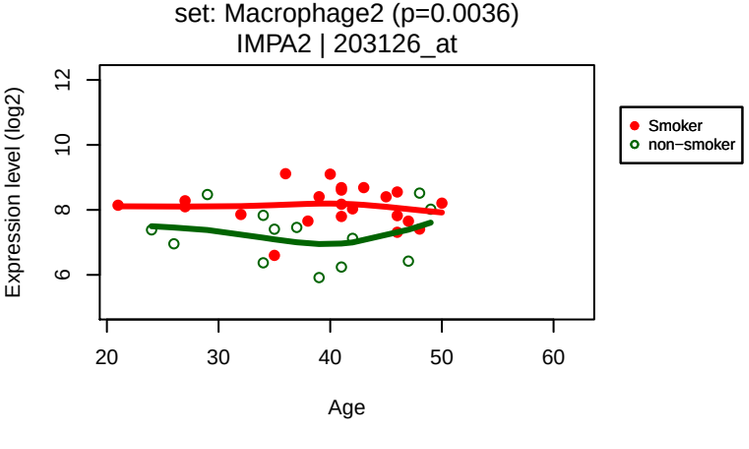

set: SmallAirway2 (p=0.0057)
IMPA2 | 203126_at
Expression level (log2)
Age
Smoker
non−smoker

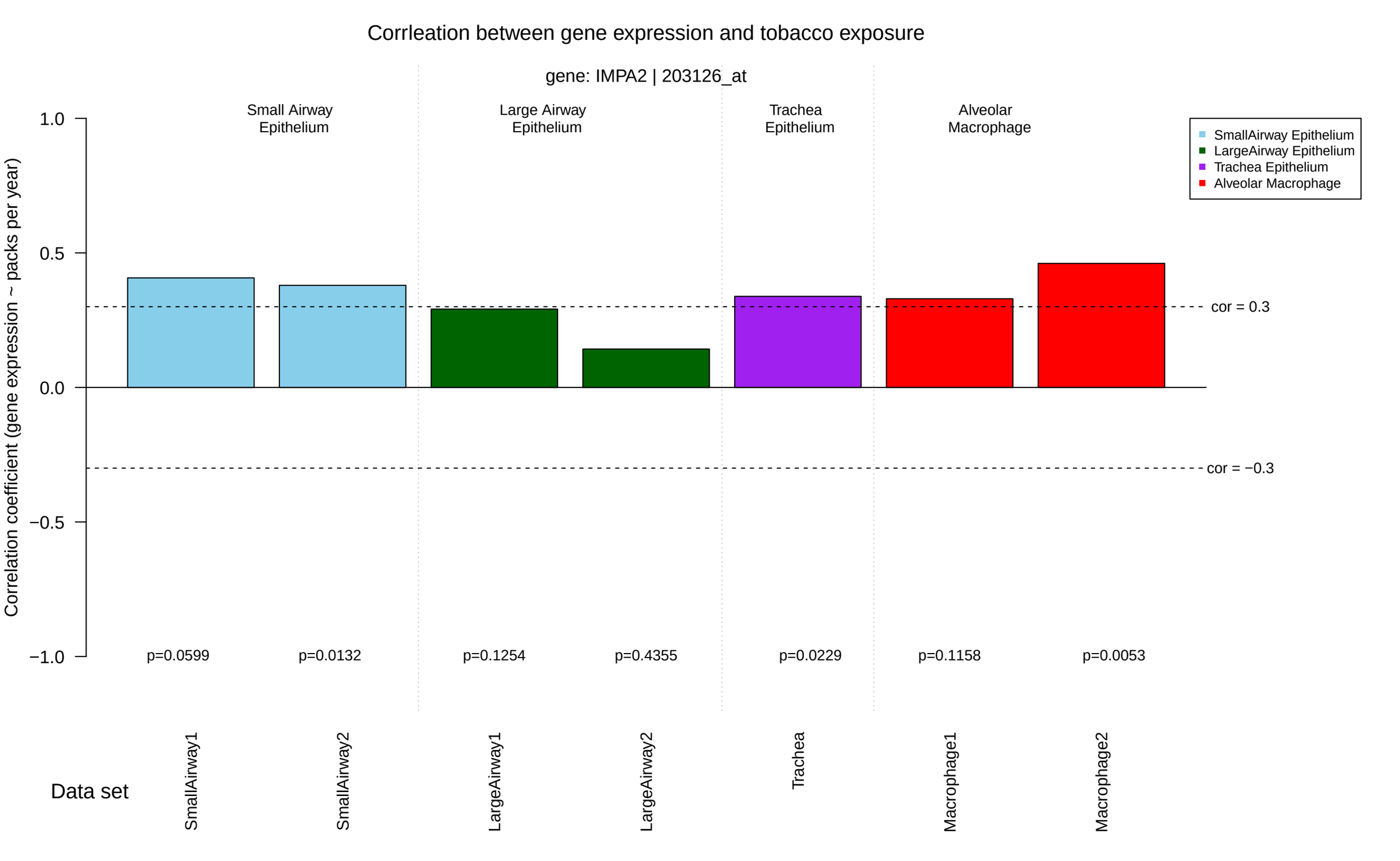

set: LargeAirway1 (p=0.1638)
IMPA2 | 203126_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0145)
IMPA2 | 203126_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0069)
IMPA2 | 203126_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0717)
IMPA2 | 203126_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0036)
IMPA2 | 203126_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IMPA2 | 203126_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0599
p=0.0132
p=0.1254
p=0.4355
p=0.0229
p=0.1158
p=0.0053
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2